Krishnaa **Vadivel** | GPU Software Engineer | High-Performance Computing

516-853-5663 • srikrishnaa.careers@gmail.com • sri-krishnaa-ece-phd
Website | GitHub: krishnaa423 | LeetCode: krishnaa42342 | Docker Hub: krishnaa42342

Profile

GPU-focused computational scientist and PhD researcher for performance-oriented systems and HPC roles. Develops Python and C++ code with CUDA, ROCm, MPI, and OpenMP, with experience in collective communication, multi-GPU parallelization, distributed debugging, and scaling analysis on national-lab HPC platforms.

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell, SQL

GPU / HPC: CUDA, ROCm, MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, collective communication

Platforms: AMD MI250X, NVIDIA A100, Frontier, Perlmutter, Linux command line, macOS, Windows, CMake

Debugging / Perf: TotalView, Linaro DDT, CUDA-aware distributed debugging, strong scaling, weak scaling, hardware-aware optimization

Machine Learning: PyTorch, PyTorch Geometric, automatic differentiation, equivariant graph neural networks, scikit-learn

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher — *2021–present*

- Developed distributed scientific software in Python and C++ with MPI, OpenMP, PETSc, and SLEPc for large linear-algebra and simulation workloads as part of an INCITE award using exascale resources on Frontier at OLCF, Perlmutter at LBNL, and related national-lab HPC systems.
- Parallelized matrix and vector workloads across multiple GPUs, using collective communication and distributed linear-algebra workflows to scale simulations across clusters.
- Ported and optimized computational workloads for AMD MI250X and NVIDIA A100 GPU systems using CUDA and ROCm, translating many-body and excited-state physics methods into scalable GPU implementations.
- Developed many-body and field-theoretic formulations of coupled electron, exciton, and phonon systems, including coherent-state and path-integral approaches for time-dependent and self-localized states.
- Used PyTorch, PyTorch Geometric, and equivariant graph neural networks to accelerate self-trapped-exciton calculations and related materials simulations, achieving roughly 100x speedups in parts of the workflow by using training and automatic differentiation to obtain additional quantities without separate expensive calculations.
- Ran strong- and weak-scaling studies and used TotalView, CUDA-aware MPI workflows, Linaro DDT, Bash, and Linux command-line tools to debug distributed applications across multi-process and multi-GPU environments.
- Packaged scientific applications in Docker containers for deployment across different computing environments.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on self-trapped exciton formation, exciton-polarons, and related excited-state materials physics.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer — *2019–2021*

- Built CUDA-based data-processing pipelines and GPU-enabled engineering tooling for large acoustic waveform datasets generated by downhole logging tools.
- Built backend tooling for database design, real-time remote tool testing, and calibration of temperature and pressure sensors.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support in-house physics and engineering studies.

FindLight

Photonics Technical Writer — *2018–2019*

- Produced technical articles and product-facing photonics content covering lasers, optics, and optoelectronic components for a specialized engineering readership.

Selected Projects

Multi-GPU Exciton-Phonon Simulation

- Developed distributed first-principles calculations for excited-state and exciton-phonon problems with CUDA, ROCm, MPI, and OpenMP, including multi-GPU matrix and vector parallelization on leadership-class GPU clusters.

Scientific ML on GPU Systems

- Developed PyTorch Geometric equivariant models for molecular-property prediction and for accelerating scientific calculations, with roughly 100x workflow speedups from combining training with automatic differentiation.

CUDA Containers for HPC Software

- Packaged scientific software in CUDA-ready Docker environments for portable deployment across local and HPC systems.

Matrix-Multiply AI Accelerator

- Built a Verilog-based matrix-multiply accelerator project for AI-oriented datapath design and hardware-software performance exploration.

Education

Yale University
PhD, Electrical Engineering — *2021–present*
Ongoing doctoral research in computational materials, many-body theory, and accelerator-enabled scientific computing.

Yale University
MPhil, Electrical Engineering — *2023*

Yale University
MS, Electrical Engineering — *2023*

Cornell University
MEng, Electrical and Computer Engineering — *2017–2018*
Included work in stochastic computing, device-oriented EE topics, and advanced design coursework.

Cornell University
BS, Electrical and Computer Engineering — *2013–2017*
Included embedded systems, robotics, and core mathematics, physics, and computing foundations.

Selected Publications

2025: de Paula, A. M., Li, S., Hou, B., Vadivel, S., Teles-Ferreira, D. C., Iudica, A., Kabacinski, P., Hosseini, H., McArthur, J., Cerullo, G., et al. (2025). Time-domain observation of ultrafast self-trapped exciton formation in lead-free double halide perovskites. *Journal of the American Chemical Society, 147*(32), 28923–28931.

Selected Conference Presentations

2026: Vadivel, S., Grande, R. D., Strubbe, D. A., & Qiu, D. Y. (2026). First-principles study of exciton-polarons and self-trapped excitons. *APS Global Physics Summit 2026*.

2025: Vadivel, S., Qiu, D. Y., Grande, R. D., & Strubbe, D. A. (2025). First-principles study of self-trapped exciton formation in double perovskites. *APS Global Physics Summit 2025*.

2024: Vadivel, S., & Qiu, D. (2024). First-principles study of self-trapped exciton formation in double perovskites using excited state forces. *APS March Meeting Abstracts 2024*, F01–007.

Selected Coursework

Mathematics: Differential Geometry; Abstract Algebra; Honors Mathematical Analysis II

Computer Science: Theoretical Computer Science; Probabilistic Machine Learning

Physics: Graduate Quantum Mechanics I–II; Solid State Physics I–II; Many-Body Quantum Physics

Electrical Engineering: Semiconductor Physics; Thin Film Science; Lasers and Optoelectronics; Integrated Photonics; VLSI

Design; Analog VLSI Design; Mathematics of Signals and Systems; Advanced Embedded Systems